

Altered Attentional Processing of Facial Expression Features in Severe Alcohol Use Disorder: An Eye-Tracking Study

Arthur Pabst¹, Zoé Bollen¹, Nicolas Masson^{2, 3}, Mado Gautier¹, Christophe Geus⁴, and Pierre Maurage¹

¹Louvain Experimental Psychopathology Research Group (LEP), Psychological Science Research Institute, UCLouvain

²Numerical Cognition Group, Psychological Science Research Institute and Neuroscience Institute, UCLouvain

³Cognitive Science and Assessment Institute, University of Luxembourg

⁴Psychiatric Unit, Cliniques Saint-Pierre d'Ottignies, Ottignies, Belgium



Social cognition impairments, and notably emotional facial expression (EFE) recognition difficulties, as well as their functional and clinical correlates, are increasingly documented in severe alcohol use disorder (SAUD). However, insights into their underlying mechanisms are lacking. Here, we tested if SAUD was associated with alterations in the attentional processing of EFEs. In a preregistered study, 40 patients with SAUD and 40 healthy controls (HCs) had to identify the emotional expression conveyed by faces while having their gaze recorded by an eye-tracker. We assessed indices of initial (first fixation locations) and later (number of fixations and dwell-time) attention with reference to regions of interest corresponding to the eyes, mouth, and nose, which carry key information for EFE recognition. We centrally found that patients had less first fixations to key facial features in general, as well as less fixations and dwell time to the eyes specifically, relative to the rest of the face, compared to controls. These effects were invariant across emotional expressions. Additional exploratory analyses revealed that patients with SAUD had a less structured viewing pattern than controls. These results offer novel, direct, evidence that patients with SAUD's socioaffective difficulties already emerge at the facial attentional processing stage, along with precisions regarding the nature and generalizability of the effects. Potential implications for the mechanistic conceptualization and treatment of social cognition difficulties in SAUD are discussed.

General Scientific Summary

This study provides evidence that patients with severe alcohol use disorder (SAUD) differ, on average, from healthy individuals in the way they attend to facial expressions: they spend proportionally less time viewing informative areas, and mostly the eyes, and have a more disorganized viewing pattern. This is important as this may partly explain why patients with SAUD have more difficulties identifying others' emotions or states of mind, and consequently present more interpersonal problems, which are known markers of poor prognosis.

Keywords: alcohol use disorder, social cognition, emotion recognition, eye-tracking, attention

Supplemental materials: <https://doi.org/10.1037/abn0000868.supp>

This article was published Online First November 13, 2023.

Pierre Maurage  <https://orcid.org/0000-0003-0197-0810>

None of the content of the present article (data, analyses, interpretation) has been disseminated earlier, neither during conferences or meetings nor through posting on a listserv or sharing on website/academic network.

Arthur Pabst (FRESH grant holder), Zoé Bollen, Mado Gautier (junior research associates), Nicolas Masson (research fellow), and Pierre Maurage (senior research associate) are funded by the Belgian Fund for Scientific Research (FRS-FNRS, Brussels, Belgium). Nicolas Masson is also funded by the National Research Fund of Luxembourg (FNR, Luxembourg). The authors declare that they have no conflicts of interest to disclose.

Arthur Pabst served as lead for conceptualization, investigation, methodology, and writing—original draft. Zoé Bollen contributed equally to resources and software and served in a supporting role for conceptualization and investigation. Nicolas Masson served as lead for data curation and validation and served in a supporting role for conceptualization, formal analysis,

methodology, and resources. Mado Gautier served in a supporting role for conceptualization, formal analysis, investigation, methodology, and validation. Christophe Geus served as lead for project administration, contributed equally to supervision, and served in a supporting role for conceptualization and investigation. Pierre Maurage served as lead for funding acquisition, resources, and supervision, contributed equally to conceptualization and project administration, and served in a supporting role for investigation and methodology. Zoé Bollen, Nicolas Masson, Mado Gautier, Christophe Geus, and Pierre Maurage contributed equally to writing—review and editing.

 The data are available at <https://osf.io/gsubj/>.

 The preregistered design is available at <https://osf.io/qd2zr>.

Correspondence concerning this article should be addressed to Pierre Maurage, Louvain Experimental Psychopathology Research Group (LEP), Psychological Science Research Institute, UCLouvain, Place du Cardinal Mercier, 10, 1348 Louvain-la-Neuve, Belgium. Email: pierre.maurage@uclouvain.be

Severe alcohol use disorder (SAUD) is associated with a broad range of social cognition difficulties, including reduced Theory of Mind and social perception/knowledge, as well as biased social attributions (F. Maurage et al., 2015; Pabst, Heeren, et al., 2020; Pabst, Maurage, et al., 2021; Pabst, Peyroux, et al., 2020; Pabst, Peyroux, et al., 2021). Among them, problems in emotional facial expression (EFE) recognition are the most firmly established: patients with SAUD identify EFE less accurately (although the recognition of positive EFE seems preserved), and tend to incorrectly perceive them as socially hostile (e.g., angry, contemptuous, disgusted; Bora & Zorlu, 2017; Castellano et al., 2015; Pabst, Heeren, et al., 2020). These EFE recognition difficulties are linked with real-life interpersonal problems (Hoffman et al., 2019), a major relapse predictor (Sliedrecht et al., 2019; Zywiak et al., 2003), as well as treatment dropout (Rupp et al., 2017). They further do not appear to naturally recover with abstinence (Rupp et al., 2021). Hence, interventions targeting this ability should be developed in SAUD. To this end, it is essential to gain insights into the mechanisms underlying these impairments.

Although their relative importance is debated, it is generally accepted that social cognition, including EFE recognition, depends both on (a) processes allowing for the extraction of relevant external signals, as well as (b) higher-order processes involved in the interpretation and integration of sensory cues with top-down information (e.g., current mental state, expectations, prior knowledge), and controlled decision-making (e.g., Adolphs, 2002; Barrett et al., 2011; Crick & Dodge, 1994; Phillips et al., 2008). In SAUD, it remains generally unclear where the difficulties originate from. However, cognitively oriented explanations have tended to focus on the latter set of processes (Uekermann & Daum, 2008), and corroborating studies have evidenced links between higher-order cognitive abilities such as episodic memory and executive functioning, and EFE recognition in SAUD (Quaglini et al., 2015). By comparison, the consideration of mechanisms that could interfere with the extraction of relevant socioaffective cues has been surprisingly limited in this group, with the notable exception of recent preliminary work suggesting that EFE recognition impairments in SAUD may also arise from difficulties in the low-level visual decoding of certain spatial frequencies (Creupelandt et al., 2019, 2022). Here, we address this issue and extend the investigation of cognitive mechanisms underlying EFE recognition impairments in SAUD toward the up-to-now neglected attentional domain. Specifically, we ask whether patients with SAUD show alterations in their attentional processing of EFE.

Indeed, several influential emotion theories contend that emotional states map onto distinct sets of characteristic physical changes, and most notoriously activations in facial muscles located around the eyes, mouth, and nose (see Barrett et al., 2019, for a review). From this perspective, attending to these specific facial areas would be particularly important for efficiently and accurately recognizing EFE. Several lines of evidence support this assertion. First, these areas capture the overwhelming majority on individuals' overt attention (i.e., eye fixations) during EFE recognition (Calvo et al., 2018; Eisenbarth & Alpers, 2011; Guo, 2012; Schurgin et al., 2014). Second, overt attention allocation to the eye region positively relates to EFE recognition (Hall et al., 2010; Sullivan et al., 2007; Vaidya et al., 2014). Third, directing participant's attention predominantly toward the eyes and mouth compared to other areas (e.g., cheeks, forehead) increases activation in face processing brain areas (Spilka et al., 2019). Fourth, experimentally occluding the eyes or

mouth reduces EFE recognition performance (Schurgin et al., 2014). Fifth, clinical groups presenting EFE recognition difficulties (i.e., patients with psychiatric, developmental, or neurological disorders) attend less to the eyes or mouth compared to their healthy counterparts (e.g., Günther et al., 2021; Kordsachia et al., 2018; Setien-Ramos et al., 2023; Toh et al., 2011; Waldthaler et al., 2019), suggesting that reduced attention to these areas affects performance.

In SAUD, impairments in attention, and specifically attentional control (P. Maurage et al., 2014) have been reported, which may impede goal-directed focus on specific areas (Schurgin et al., 2014). SAUD is further linked with modifications in attentional processing event-related potentials (N2b/P3a) when viewing EFE (P. Maurage et al., 2008). Additionally, patients with SAUD present reduced gray matter volume (Le Berre et al., 2013; Makris et al., 2008; Trick et al., 2014) and functional activity during EFE processing (Marinkovic et al., 2009; O'Daly et al., 2012; Salloum et al., 2007) in the amygdala and frontal cortex, which are involved in guiding gaze toward informative areas, and notably the eyes (Adolphs et al., 2005; Gamer et al., 2013; Kennedy & Adolphs, 2010; Vaidya & Fellows, 2019; Wolf et al., 2014), albeit possibly via different routes (i.e., initial direction of attention for the amygdala; later attentional engagement for the frontal cortex). Finally, SAUD is associated with heightened sensitivity to socially threatening or hostile cues, including angry and disgusted EFE (P. Maurage et al., 2012; Pabst, Heeren, et al., 2020; Pabst, Peyroux, et al., 2020; Park et al., 2015, 2017), which may either interfere with efficient attention allocation or motivate patients to direct their attention away from informative areas of these stimuli to avoid distress (as described in social anxiety; Horley et al., 2004). In summary, an emerging literature suggests that SAUD may also be marked by altered facial attention when processing EFE. However, the evidence is highly preliminary and relies on decontextualized or indirect (neurophysiological) indices, thus highlighting the need for confirmation using a direct measure of attention. Moreover, the available studies do not offer definitive insights as to (a) which facial areas (eyes, mouth, nose) are abnormally attended to (b) whether difficulties arise at initial (first fixations) and/or later (dwell-time, total fixations) stages of attention, nor (c) whether they are emotion-specific or general. Addressing these limitations is crucial to understand the etiology of and intervene on EFE processing difficulties in SAUD.

Consequently, using an eye-tracking paradigm, we conducted the first direct test of the general hypothesis that patients with SAUD show reduced attention to informative facial areas during EFE recognition compared to HCs. Additionally, we determined which facial areas (eyes, mouth, nose) are concerned. We further disentangled which stages of attention (initial vs. later) are affected, and whether effects are emotion-specific (e.g., only observed for angry, or disgusted EFE). Secondarily, in line with previous behavioral results (Bora & Zorlu, 2017; Freeman et al., 2018; P. Maurage et al., 2021), we also hypothesized that patients with SAUD would be less accurate in recognizing EFE, especially low-intensity hostile ones (i.e., anger, contempt, disgust), and would more often incorrectly attribute hostile emotions than HCs.

In complement to traditional indices of attentional focus during EFE processing (i.e., "what" is attended to), we explored gaze entropy, or the predictability of gaze patterns (Krejtz et al., 2015; Shiferaw et al., 2019). When processing EFE, lower gaze entropy reflects a more predictable gaze pattern indicating that attentional exploration is more influenced by top-down emotional priors, and

hence more structured and strategic (Cuve et al., 2021; Gehrer et al., 2018). This measure offers valuable additional insights regarding “how” patients with SAUD attend to EFE, and notably evaluates whether SAUD is associated with a less strategic, more random facial exploration. Finally, we explored possible associations between indices of attention and clinical variables and investigated potential confounding effects of psychopathology symptoms.

Method

Participants (Table 1)

We recruited 40 patients with a *Diagnostic and Statistical Manual of Mental Disorders*—5th ed. (American Psychiatric Association, 2013) diagnosis of SAUD undergoing detoxification in specialized Belgian hospitals (Cliniques Saint-Pierre Ottignies, Le Beau-Vallon, Centre Hospitalier Le Domaine). Patients had all abstained from alcohol for at least 10 days. Prior to hospital admission, a trained psychiatrist interviewed them to ensure the absence of major physical, neurological, or additional psychiatric disorders. Included patients had no other lifetime neurological or psychiatric diagnoses besides depression, anxiety, or nicotine use disorder, and reported SAUD as their current primary problem. We also recruited 40 age- and gender-matched HCs via social media, University participant pools, and flyer advertisements. They reported no lifetime neurological or psychiatric disorders except nicotine use disorder and had no family history of SAUD. Additionally, they reported drinking less than 10 units per week and never more than 3 units per day and had an Alcohol Use Disorder Identification Test (AUDIT; Babor et al., 2001) score below 8. HCs were compensated 5 euros per 30 min of participation. All participants had normal or corrected-to-normal vision. The study protocol was approved by the biomedical ethical committee of the local faculty of medicine (UCLouvain) and complied with the Declaration of Helsinki. All participants provided informed written consent.

We determined the sample size required to test our main eye-tracking hypotheses by an a priori power analysis conducted on

G*Power v3.1.9.4 (see Gehrer et al., 2019, for a similar approach). It revealed a necessary sample of 13 participants per group to detect a mixed Group $\times$ Regions of Interest (ROI) $\times$ Emotional Expression interaction (15 measurements) and 36 participants per group to detect a mixed group $\times$ ROI (three measurements) interaction, assuming a small-to-medium ($f = 0.175$) effect size and a desired power of .90 with an α of .05. We recruited slightly more participants per group ($n = 40$) than required to detect the least powerful interaction ($n = 36$) to counter unforeseeable data loss and maximize power for exploratory analyses.

Stimuli

We used a total of 160 colored photographs depicting eight Caucasian actors (four females) expressing anger, contempt, disgust, fear, and sadness from the Radboud database (Langner et al., 2010). The actors were trained to express these emotions by executing the facial movements described in the Facial Action Coding System (Ekman et al., 2002; see Langner et al., 2010, Figure 2 for a list of the specific facial movements performed for each emotion). We chose this database because the location of facial features is standardized within actors and because it comprises contemptuous expressions which are abnormally recognized in SAUD (P. Maurage et al., 2021; Philippot et al., 1999). Based on meta-analytical evidence suggesting a relatively preserved recognition of positive EFE in SAUD (Bora & Zorlu, 2017), we decided to minimize testing time by including only negative emotions. We equalized all photographs for luminance, and on-screen dimensions were 21.2×18.3 cm (height $\times$ length). To reflect the facts that daily-encountered EFE varies in intensity, and that decoding impairments in SAUD may be stronger for subtler stimuli (P. Maurage et al., 2021), we created morphs of varying intensities by combining each emotion of each actor with the neutral expression of the same actor using Fantamorph 5.0 (Abrosoft; <https://www.fantomorph.com>). Each emotion was presented twice at 0%, 10%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 80%, 90%, and 100% intensity. We included the zero intensity to investigate response biases when no emotion was expressed (i.e., neutral expression). Each actor was presented four times for each emotion, at different intensity levels. Stimuli were divided into two blocks of 80 photographs, each comprising one set of four identities (two females in each block) to allow for a break. All participants viewed the two blocks. Block order was counterbalanced and within-block stimuli presentation was randomized.

Task

Participants were instructed to identify the emotion expressed by each photograph as accurately as possible, without thinking too long about their answer. A six-trial training phase with stimuli not included in the experimental blocks preceded the task. Each trial began with a fixation cross acting as a drift check randomly appearing on the left or right side of the screen. This was done to prevent artificial orientation of participants' gaze toward the center of the photographs before their appearance. After registration of a 1,000 ms fixation on the fixation cross, the photograph immediately appeared in the center of the screen for 3,000 ms. Following this, a response display simultaneously comprising the labels “anger,” “contempt,” “disgust,” “fear,” and “sadness,” in random order, was

Table 1
Sociodemographic, Psychopathological, and Alcohol-Related Measures among Participants With SAUD and HC Participants: Mean (SD)

Variable	SAUD ($n = 40$)	HC ($n = 40$)
Age	45.28 (9.55)	47.33 (11.68)
Gender, N (% females)	12 (30)	17 (42.5)
Years of completed education years**	13.40 (2.18)	15.73 (2.21)
BDI**	9.89 (8.10) ^b	2.68 (2.83)
LSAS	45.21 (22.24) ^a	37.93 (26.61)
STAI trait**	48.93 (12.79) ^b	35.78 (10.47)
STAI state*	32.58 (12.32) ^a	27.02 (7.57)
AUDIT**	30.53 (5.48) ^a	2.95 (1.50)
Daily consumption (standard units)	18.86 (12.85)	
Duration of SAUD (years)	8.56 (7.34)	
Number of previous detoxifications	2.38 (4.07)	

Note. SAUD = severe alcohol use disorder; HC = healthy control; BDI = Beck Depression Inventory; LSAS = Liebowitz Social Anxiety Scale; STAI = State-and-Trait Anxiety Inventory; AUDIT = Alcohol Use Disorder Identification Test.

^a One missing value. ^b Two missing values.

* $p < .05$. ** $p < .001$.

presented until an answer was selected via mouse-click. We did not include a “neutral” label despite the presence of zero-intensity expressions to prevent default answers when faced with uncertainty and to facilitate the observation of biases.

Setting, Apparatus, and Software

The experiment took place in a quiet, dimly lit room, with participants seated on an adjustable chair. Stimuli appeared on an Asus laptop with a 17.3" Full HD screen (1,080 × 1,920p, 120 Hz refresh rate), located 60 cm away from participants' eyes. We used Opensesame (Mathôt et al., 2012) for stimuli presentation and eye-tracking synchronization. We recorded eye movements from the right eye using the state-of-the-art Eye-link Portable Duo (SR Research, Canada; 1,000 Hz sampling rate; average accuracy range 0.25°–0.5°, gaze tracking range of 32° horizontally, 25° vertically) and we parsed gaze events into fixations using the EyeLink DataViewer software. Before training and before each experimental block, participants underwent a standard nine-point calibration.

Eye-Movement Processing

We extracted participants' fixations (as defined by the EyeLink DataViewer's default settings) on the face photographs. We then merged fixations separated by less than 1° of visual angle and deleted fixations shorter than 100 ms (Manor & Gordon, 2003). We removed fixations starting before the onset of the photographs and trimmed the duration of fixations lasting until after their disappearance to focus exclusively on photograph-contingent gaze behavior. In order to analyze participants' fixations on specific facial features, we further defined ROI tailored to each actor (facial features were aligned within actors). These consisted of large rectangles superimposed on each eye, the mouth, the nose, and the whole face (excluding background, shirt, and hair). We selected these ROIs based on theory and previous research indicating that these facial areas are most often attended to and/or are most informative during emotion recognition (Calvo et al., 2018; Eisenbarth & Alpers, 2011; Guo, 2012; Schurgin et al., 2014).

Psychopathology Questionnaire Measures

We measured depression symptoms with the Beck Depression Inventory (BDI; Beck & Steer, 1987), anxiety symptoms with the State-and-Trait Anxiety Inventory (STAI; Spielberger & Gorsuch, 1983), and social anxiety symptoms with the Liebowitz Social Anxiety Scale (LSAS; Liebowitz, 1987).

Analyses

Data Exclusion

For eye-tracking analyses, we excluded data from 72 trials in the SAUD sample (1.12% of SAUD trials) and 8 trials in the HC sample (0.12% of HC trials), for which more than 50% of fixations occurred outside the face ROI.

Main Analyses: Eye-Tracking Data

To index initial attention to facial features, we computed for each participant and emotion, the proportion of trials in which the first fixation

occurred on the eyes (left and right combined), mouth, or nose ROI. To index later attention to facial features, we computed, for each participant and emotion the mean proportion of fixations and dwell-time (sum of fixation durations) on the eyes, mouth, and nose, relative to the face ROI. We preferred proportions over absolute attention indices, as the latter may be affected by patients with SAUD's reduced visual processing speed (e.g., Bertera & Parsons, 1978). We submitted the initial attention allocation index to a 2 (group, between) × 3 (ROI, within) × 5 (emotional expression, within) mixed analysis of variance (ANOVA). Because groups unexpectedly differed in completed education years (see Results section), and preliminary correlation analyses showed significant or almost significant associations between education and proportions of total fixations and dwell time indices (education was not associated with other eye-tracking indices), we deviated from our preregistered analysis plan and submitted both indices of overall attention to 2 (group, between) × 3 (ROI, within) × 5 (emotional expression, within) mixed analyses of covariance (ANCOVAs) with education (centered) as a covariate, instead of ANOVAs. We excluded zero-intensity trials from these analyses to maintain a meaningful emotional expression factor.

Secondary Analyses: Behavioral Emotion Recognition Data

To index EFE recognition accuracy, we computed for each participant, emotional expression, and intensity category, defined as low (10%–35%), medium (40%–60%), and high (65%–100%), the proportion of trials with correct emotion recognition (0%–intensity trials were excluded from this analysis). To index biases in emotion recognition, we computed, for each participant and emotion expression displayed (0%–intensity trials were included and considered to display a neutral expression), the proportion of incorrect trials in which a given emotional expression was selected. As education was significantly associated with emotion recognition performance indices, the proportions of trials with correct emotion recognitions were submitted to a 2 (group, between) × 5 (emotional expression, within) × 3 (intensity class) mixed ANCOVA with education as a covariate, instead of ANOVA. Proportions of incorrect emotion selections were submitted to a 2 (group, between) × 5 (selected emotion, within) mixed ANOVA.

We applied Greenhouse–Geisser corrections where appropriate. We performed all analyses on R (R Core Team, 2019), supplemented with the “afex” (Singmann et al., 2021) and “emmeans” (Lenth et al., 2021) packages. We estimated the sizes of relevant effects using partial eta squared and Cohen's *d*.

Additional Exploratory Analyses

Gaze Transition Entropy. Gaze transition entropy reflects the extent to which ROI transition destinations are predictable given current fixation locations, and thus indexes how much visual exploration is structured, strategic (influenced by top-down factors) versus random. Based on matrices of fixation transition frequencies from each ROI (eyes, mouth, nose, rest of the face) to the same or a different ROI (fixations occurring outside the face were excluded) obtained for each participant and each emotional expression (zero-intensity trials excluded), we computed gaze transition entropy using the formula detailed by Shiferaw et al. (2019):

$$H(x) = - \sum_{i=1}^n p_i \sum_{j=1}^n p(i|j) \log_2 p(i|j)$$

H is the uncertainty of x given its prior state, p_i is the observed absolute probability of viewing the i th ROI (max $n = 4$), and $p(i|j)$ is the probability of transitioning from i to j . Scores were normalized by maximum entropy, defined as the binary logarithm of the number of different ROIs (i.e., 4) that could characterize the location of gaze. Higher gaze transition entropy indicates more random gaze transitions. We submitted these scores to a 2 (group, between) by 5 (emotional expression, within) mixed ANOVA.

Links Between Eye-Tracking Indices and Psychopathology Symptoms and EFE Recognition. We used Pearson correlations to explore associations between eye-tracking indices differing between groups and psychopathology symptoms differing between groups, and investigated the confounding effect of psychopathological symptoms found to be associated using ANCOVA. We also used Pearson correlations to explore the links between these eye-tracking indices and EFE recognition. Significant associations would provide empirical support to the view that altered attention to relevant facial areas plays a role in EFE recognition impairments.

Transparency and Openness

We preregistered the hypotheses, design, sample size rationale, and main analysis plan on the Open Science Framework (Pabst, Maurage, et al., 2021; Pabst, Peyroux, et al., 2021). We present departures from this original plan in the corresponding sections below. Exploratory analyses without preregistered hypotheses are presented as such. The data (anonymized and without age and gender data) and R code used for our analyses can also be found on the Open Science Framework page.

Results

Sample Characteristics (Table 1)

There were no group differences in age, gender and social anxiety (LSAS) symptoms, but patients with SAUD reported less education years than HC, together with more elevated depression (BDI) and state/trait anxiety (STAI) symptoms. Patients with SAUD also logically reported higher AUDIT scores.

Eye-Tracking Main Results (Figure 1)¹

Initial Attention Allocation to Facial Features

The ANOVA on proportions of first fixations revealed a main effect of group, $F(1, 78) = 17.51, p < .001, \eta_p^2 = .18$, with lower proportions of first fixations occurring on any ROI (proportions averaged across ROIs) in SAUD ($M = 0.19, SE = 0.007$) than HC ($M = 0.23, SE = 0.007$), meaning that patients made proportionally less first fixations toward any ROI than HC. The group $\times$ ROI, $F(1.22, 94.89) = 1.20, p = .286, \eta_p^2 = .02$ and the Group $\times$ Emotional Expression did not reach significance, $F(3.89, 303.04) = 1.06, p = .377, \eta_p^2 = .01$, but there was a significant Group $\times$ ROI $\times$ Emotional Expression interaction, $F(6.18, 481.70) = 2.29, p = .033, \eta_p^2 = .03$. Follow-up uncorrected ANOVAs showed that this effect was due to Group $\times$ Emotional Expression interactions selectively emerging for proportions of first fixations on the mouth ($p = .021$) and nose ($p = .017$). No group $\times$ ROI interaction emerged at any emotional expression level (all $ps > .18$). Additional follow-up comparisons showed that patients with SAUD made proportionally more first fixations toward the mouth

when faces expressed fear or disgust compared to anger and sadness ($ps < .044$). No difference between emotional expressions was observed in HC (all $ps > .84$). On the opposite, regarding the nose, an emotional expression effect was found in HC but not in SAUD (all $ps > .32$), since HC made proportionally more first fixations toward the nose when faces expressed disgust compared to anger ($p < .012$) and fear ($p = .05$). Despite these complex group-specific emotional expression effects, there were no uncorrected significant between-group effects on the proportion of first fixations on the mouth or nose for any emotional expression (all $ps > .13$). Hence, we did not consider this three-way interaction further.

Later Attention to Facial Features

The ANCOVA on proportions of total fixations revealed again a main effect of group, $F(1, 77) = 4.48, p = .037, \eta_p^2 = .06$, with lower proportions of fixations occurring on any ROI (proportions averaged across ROIs) in SAUD ($M_{\text{education adjusted (EA)}} = 0.27, SE = 0.003$) than HC ($M_{\text{EA}} = 0.28, SE = 0.003$). There was also a significant Group $\times$ ROI interaction, $F(1.36, 104.64) = 3.80, p = .041, \eta_p^2 = .05$. Uncorrected follow-up between-subjects comparisons showed that this interaction was due to patients with SAUD having lower proportions of total fixations on the eyes, SAUD: $M_{\text{EA}} = 0.39, SE = 0.03$; HC: $M_{\text{EA}} = 0.49, SE = 0.03$; $t(77) = 2.26, p = .026, d = 0.52$ and higher (although only at a trend level) proportions of fixations on the nose, SAUD: $M_{\text{EA}} = 0.20, SE = 0.02$; HC: $M_{\text{EA}} = 0.15, SE = 0.02$; $t(77) = 1.86, p = .066, d = 0.42$ than HC, with no difference regarding proportions of fixations on the mouth, SAUD: $M_{\text{EA}} = 0.22, SE = 0.02$; HC: $M_{\text{EA}} = 0.19, SE = 0.02$; $t(77) = 0.87, p = .387, d = 0.20$. The Group $\times$ Emotional Expression, $F(3.75, 288.62) = 0.29, p = .873, \eta_p^2 = .004$, and the Group $\times$ ROI $\times$ Emotional Expression interactions were not significant, $F(6.42, 493.97) = 1.42, p = .201, \eta_p^2 = .02$.

The ANCOVA on proportions of dwell-time showed no main effect of group, $F(1, 77) = 2.65, p = .107, \eta_p^2 = .03$, but again revealed a significant Group $\times$ ROI interaction, $F(1.42, 109.52) = 3.67, p = .043, \eta_p^2 = .05$. Uncorrected follow-up between-group comparisons again showed that this interaction was due to patients with SAUD having lower proportions of dwell-time on the eyes, SAUD: $M_{\text{EA}} = 0.41, SE = 0.03$; HC: $M_{\text{EA}} = 0.51, SE = 0.03$; $t(77) = 2.23, p = .028, d = 0.51$ and higher (although only at a trend level) proportions of dwell-time on the nose, SAUD: $M_{\text{EA}} = 0.18, SE = 0.02$; HC: $M_{\text{EA}} = 0.13, SE = 0.02$; $t(77) = 1.96, p = .054, d = 0.45$, than HC, with no difference regarding proportions of dwell-time on the mouth, SAUD: $M_{\text{EA}} = 0.24, SE = 0.02$; HC: $M_{\text{EA}} = 0.21, SE = 0.02$; $t(77) = 0.94, p = .350, d = 0.21$. The Group $\times$ Emotional Expression, $F(3.75, 288.75) = 0.60, p = .652, \eta_p^2 = .01$, and Group $\times$ ROI $\times$ Emotional Expression interactions were not significant, $F(6.54, 503.69) = 1.49, p = .175, \eta_p^2 = .02$.

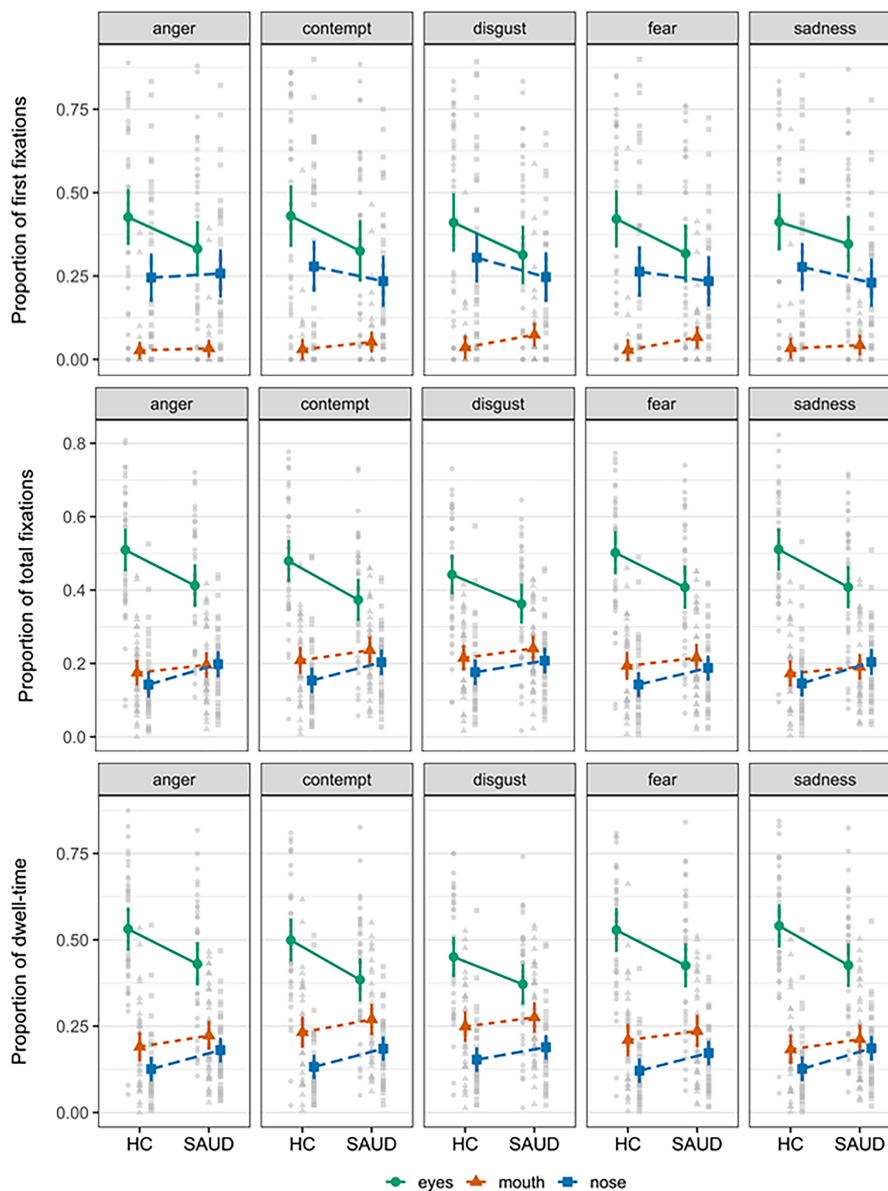
Eye-Tracking Exploratory Results: Gaze Transition Entropy

The ANOVA on gaze transition entropy revealed a main effect of group, $F(1, 78) = 4.21, p = .044, \eta_p^2 = .05$, with higher scores (i.e.,

¹ We only report relevant effects involving the group factor, as it constitutes our main focus.

Figure 1

Mean Proportions of First Fixations, Total Fixations, and Dwell-Time Occurring on the Eyes, Mouth, and Nose, Relative to the Rest of the Face, as a Function of Group and Emotional Expression



Note. HC = healthy control; SAUD = severe alcohol use disorder. See the online article for the color version of this figure.

less strategic exploration) in patients with SAUD ($M = 0.83$, $SE = 0.02$) than HC ($M = 0.78$, $SE = 0.02$). The Group $\times$ Emotional Expression interaction was not significant, $F(3.37, 262.55) = 1.44$, $p = .229$, $\eta_p^2 = .02$.

Behavioral Main Results (See the Online Supplemental Materials for Descriptive Statistics)

The ANCOVA on proportions of correct emotion recognition trials revealed that the main effect of group, $F(1, 77) = 0.52$, $p = .47$, $\eta_p^2 = .01$, the Group $\times$ Emotional Expression, $F(3.51, 269.90) =$

0.85 , $p = .481$, $\eta_p^2 = .01$, the Group $\times$ Intensity Class, $F(1.87, 143.62) = 1.57$, $p = .213$, $\eta_p^2 = .02$, and the Group $\times$ Emotional Expression $\times$ Intensity Class interactions, $F(5.87, 451.90) = 0.48$, $p = .822$, $\eta_p^2 = .01$ were all nonsignificant. Similarly, the ANOVA on the proportions of incorrect emotion selections revealed that the relevant Group $\times$ Selected Emotion, $F(2.32, 181.33) = 0.17$, $p = .875$, $\eta_p^2 = .002$ was not significant. An additional exploratory (nonpreregistered) analysis conducted only on zero-intensity neutral trials again revealed a Nonsignificant Group $\times$ Selected Emotion interaction, $F(1.85, 144.14) = 0.43$, $p = .638$, $\eta_p^2 = .005$.

Additional Exploratory Results

Links Between Eye-Tracking Indices and Psychopathology

Given the above-presented results, we explored links between depression, state anxiety, and trait-anxiety symptoms on one hand, and proportions of first and total fixations occurring on any ROIs (averaged across ROI), proportions of fixations and dwell-time on the eyes, and gaze transition entropy on the other hand. No significant correlation emerged regarding proportions of first fixations, proportions of dwell-time, or gaze transition entropy in either group (all $ps > .34$). There were however correlations between proportions of total fixations averaged across ROIs and depression ($r = -.36$, $p = .025$) and, at a trend level, trait anxiety ($r = -.27$, $p = .088$) in HC. Adding depression and trait anxiety symptoms as covariates in the previously described ANCOVA on proportions of total fixations did not alter its general conclusions: the group, $F(1, 73) = 3.28$, $p = .074$, $\eta_p^2 = .04$, and Group $\times$ ROI, $F(1.36, 98.95) = 3.38$, $p = .056$, $\eta_p^2 = .04$ effects remained marginally significant and the follow-up between-group comparisons confirmed the lower proportions of fixations on the eyes, $t(73) = 2.03$, $p = .046$, $d = 0.48$ and higher proportion of fixations on the nose, $t(73) = 2.17$, $p = .033$, $d = 0.51$ in the SAUD group compared to HC.

Links Between Eye-Tracking Indices and EFE Recognition

We explored the links between percentages of first and total fixations on relevant areas (averaged across ROI), percentages of dwell time and total fixations on the eyes, and gaze transition entropy on the one hand, and EFE recognition accuracy indices for each emotion on the other hand, both in the full sample and in patients with SAUD only. In the full sample, fear recognition correlated positively with the percentage of first ($r = .36$, $p = .0009$) and total ($r = .22$, $p = .048$) fixations on relevant facial areas, as well as with the percentages of dwell-time ($r = .20$, $p = .07$) and total fixations on the eyes ($r = .22$, $p = .049$), all other ps were $>.09$. The pattern of results was similar in patients with SAUD only: fear recognition correlated with the percentages of dwell-time ($r = .30$, $p = .06$) and total fixations ($r = .32$, $p = .04$) on the eyes. However, the correlation with the percentages of first ($r = .22$, $p = .18$) and total ($r = .09$, $p = .58$) fixations on relevant facial areas became nonsignificant, all other ps were $>.19$.

Discussion

We provided the first direct investigation of attention allocation during EFE recognition in SAUD. Our main findings support the hypothesis that SAUD is associated with reduced attention to key facial features, and predominantly the eyes, and further allow for a characterization of this effect.

This reduced attention concerned both initial and later attention indices. Patients with SAUD's first fixations occurred proportionally less often on relevant ROIs compared to HC. Although the effect appears most consistent for the eyes, the absence of a significant interaction precludes the conclusion that this result is due to reduced first fixations toward this region specifically and rather suggests that patients with SAUD have a globally reduced initial tendency to orient their attention to informative facial features. Patients with SAUD also displayed reduced proportions of total fixations and dwell-time compared to HC, on the eye region but not on the mouth/nose. This

suggests that the classical attentional focus on the eye region, repeatedly observed earlier (e.g., Schurigin et al., 2014) and confirmed in our HC group, is attenuated in SAUD. Finally, the effects did not significantly differ by emotional expression, indicating that they are not explained by differences in the attentional processing of specific (e.g., hostile) EFE. In summary, patients with SAUD showed reduced early, automatic attention to informative facial areas in general, and reduced later attention to the eyes specifically, regardless of the expressed emotion. Importantly, these findings were independent of, or persisted after controlling for, education and relevant psychopathological symptoms.

These results indicate that attentional impairments in SAUD (P. Maurage et al., 2014) extend to socioaffective stimuli, confirming previous indirect electrophysiological results showing modified attentional processing of EFE (P. Maurage et al., 2008). This reinforces proposals to conceptualize SAUD, not only as a "cold" cognitive, but also as a socioaffective disorder (e.g., P. Maurage et al., 2021). Indeed, it suggests that socioaffective information processing alterations in SAUD can be traced back to a mechanism as fundamental to social cognition and interpersonal interactions (Cañigueral & Hamilton, 2019; Itier & Batty, 2009) as attentional allocation to key facial features such as the eyes. Interestingly, the magnitude of the reported effects is comparable to those found in other psychiatric conditions primarily marked by social impairments, such as autism (Setien-Ramos et al., 2023) and social anxiety disorder (Günther et al., 2021). Besides further stressing the interpersonal dimension of SAUD, these observations also highlight the transdiagnostic relevance of altered facial attention. Most importantly, the current findings add to efforts aimed at identifying the cognitive mechanisms underlying EFE recognition difficulties in SAUD, by emphasizing altered attention to key facial areas as a crucial factor potentially preventing the efficient extraction of relevant facial information, and hence reducing performance.

The simultaneous reduction of initial and later attention to central facial features can be discussed in light of dominant dual-process models viewing addiction as a combination of increased sensitization to the substance, and reduced cognitive control (Koob & Volkow, 2010; Stacy & Wiers, 2010). On the one hand, reduced initial attention to key facial areas in general may reflect reduced sensitivity to social stimuli, following the hijacking of the salience systems by substance-related cues, at the expense of alternative, naturally valued and attention-grabbing social stimuli (Kalivas & Volkow, 2005; Volkow et al., 2011). On the other hand, reduced controlled attention to the eyes may result from an impaired ability to mobilize cognitive resources to voluntarily orient attention toward goal-relevant areas (P. Maurage et al., 2014). This interpretation is consistent with the contention that initial fixations reflect rather exogenous stimulus-based, and later fixations rather endogenous controlled, attention, although both processes likely influence both indices to some extent (e.g., Nummenmaa et al., 2006). Further research is required to clarify if and how social impairments in SAUD can be integrated in dual-process models. Additionally, this combined alteration suggests that structural and functional impairments found in both limbic and frontal brain areas might be involved in social cognition deficits in SAUD (Le Berre et al., 2013; Makris et al., 2008; Marinkovic et al., 2009; O'Daly et al., 2012; Salloum et al., 2007; Trick et al., 2014), particularly through their purported role in initial and later attention to relevant facial areas, respectively (Adolphs et al., 2005; Gamer et al., 2013; Kennedy & Adolphs, 2010; Vaidya & Fellows, 2019; Wolf

et al., 2014). Studies combining eye-tracking and neuroimaging techniques in SAUD are necessary to confirm this proposal. Finally, the absence of modulation by emotion indicates that the effects are not driven by emotion-specific mechanisms such as increased sensitivity to socially hostile stimuli (Pabst, Heeren, et al., 2020; Pabst, Peyroux, et al., 2020).

Our exploratory eye-tracking analyses additionally revealed that patients with SAUD also differ from HC in the structure of their attention allocation to EFE. Their increased gaze transition entropy indicates that their attentional exploration of EFE is less structured and strategic than HC's (Cuve et al., 2021; Krejtz et al., 2015; Shiferaw et al., 2019). This may again be attributable to patients' reduced attentional control (P. Maurage et al., 2014). Alternatively, within a predictive coding framework in which eye-movements are informed by, and constitute means to test, top-down sensory predictions (Friston et al., 2012), increased gaze entropy during EFE may represent novel evidence of reduced influence of emotional predictions in patients with SAUD (Cuve et al., 2021). Additional research is needed to pinpoint the mechanisms underlying increased gaze transition entropy during EFE recognition in SAUD.

Despite modified attentional processing, we found no group differences on EFE recognition indices. This contradicts previous findings (Bora & Zorlu, 2017; Castellano et al., 2015; Freeman et al., 2018; P. Maurage et al., 2021), and apparently challenges the proposed relationship between abnormal attention allocation and EFE recognition difficulties in SAUD. However, methodological aspects may have reduced the task's sensitivity to behavioral differences in EFE recognition performance. First, to optimize our task for our primary eye-tracking measures, we fixed EFE exposure time at 3,000 ms before presenting response options. Response speed could hence not be emphasized and was not recorded. Most previous EFE categorization paradigms presented stimuli for shorter durations (Freeman et al., 2018; Kornreich et al., 2013, 2016), allowed for responses as soon as stimuli appeared (e.g., Carmona-Perera et al., 2014; P. Maurage et al., 2021), and/or encouraged participants to answer as fast as possible (Foisy et al., 2007; Kornreich et al., 2016). We therefore allowed for increased EFE processing time compared to previous studies and did not collect subtler performance indicators such as reaction times, potentially obscuring group differences related to patient's reduced visual processing speed (Bertera & Parsons, 1978). Second, we focused on a subset of five, negative, EFE whereas other paradigms typically entail at least six EFE, including happiness and/or surprise (e.g., Cox et al., 2018; Kornreich et al., 2013; P. Maurage et al., 2021), which may have reduced the cognitive demand imposed on participants. Third, we did not allow for neutral responses, preventing spontaneous errors related to impressions of total absence of emotion, which may be expected in patients with SAUD in light of their higher detection thresholds (D'Hondt et al., 2015; P. Maurage et al., 2021). Fourth, with HC accurately recognizing 74.9% of EFE (72.3% in patients with SAUD), our task was easier (partly because of the abovementioned points), than comparable ones yielding group differences (Freeman et al., 2018; Kornreich et al., 2013; P. Maurage et al., 2021). For these reasons, and although our behavioral findings are valuable as they point to boundary conditions for EFE recognition impairments in SAUD, and because such null results will be important to integrate in future meta-analyses, we argue that the null findings on EFE recognition in the presence of modified attentional processing do not imply functional independence. Indeed, the link between attention to key facial areas and EFE recognition is well

established (Hall et al., 2010; Schurgin et al., 2014; Spilka et al., 2019; Sullivan et al., 2007; Vaidya et al., 2014). Furthermore, and despite our EFE recognition task's reduced sensitivity and our limited sample size, our complementary analyses revealed that fear recognition was associated with reduced first and total fixations to relevant face areas in the whole sample, and with reduced dwell time and fixations to the eyes in both the whole sample and the subsample of patients with SAUD. Although such exploratory and uncorrected results must be interpreted cautiously, they align with previous studies showing that attention to key facial features, and notably the eyes, are especially relevant for the recognition of fear (Eisenbarth & Alpers, 2011; Schurgin et al., 2014; Vaidya et al., 2014), which is principally characterized by a widening of the eyes. Crucially, the joint significance of the links between SAUD and altered facial attention, and between altered facial attention and EFE recognition performance, offers direct empirical support to the view that reduced attention to key facial areas actively contributes to impaired EFE recognition in SAUD specifically (Kornreich, 2017; MacKinnon et al., 2002), underscoring the clinical relevance of the current findings. We urge for replications using more sensitive tasks and experimental manipulations of attention using visual cues (Spilka et al., 2019) or verbal instructions (Wolf et al., 2016) during EFE processing to confirm that reduced attention to informative facial features is a mechanism in EFE recognition difficulties in SAUD.

Importantly, patients' reduced attention to the eye region, being independent of emotional content, may have clinically relevant consequences beyond impairing EFE identification. The eyes carry additional information regarding the source and social significance of individuals' expressions (Roelofs et al., 2010; Sander et al., 2007). For example, angry faces looking toward perceivers' eyes versus looking elsewhere have drastically different personal implications. Failing to take this information into account may lead patients with SAUD to erroneously interpret negative expressions as being self-directed and hostile (Pabst, Heeren, et al., 2020; Pabst, Peyroux, et al., 2020). Moreover, looking at the eyes provides information about others' non-emotional mental states such as intentions, current attentional focus, or desire to communicate and establish intimacy, and also logically constitutes a valuable signal facilitating others' access to our own (emotional) mental states (Cañigueral & Hamilton, 2019; Itier & Batty, 2009). This latter aspect is particularly relevant in clinical settings where practitioners' poor understanding of patients may deter the therapeutic alliance and reduce the adherence to and effectiveness of interventions. Hence, our findings shed light on a potential mechanism impacting the ability to understand and be understood by others in SAUD, and thus participating in social cognition and interpersonal functioning impairments more broadly.

Our work comprises limitations. First, our framework was cognitively oriented and we focused on but one potential mechanism of EFE recognition difficulties in SAUD. Alterations in higher-order processes such as social memory, expectations, or decision-making, as well as in noncognitive processes such as emotional reactivity or motor facial mimicry may additionally be involved. It will be important for future work to further investigate these processes and their relationships in order for an integrative model to emerge. Second, in line with previous behavioral EFE recognition studies in SAUD, we employed static stimuli lacking ecological validity, and we focused on Caucasian faces, hampering the investigation of cross-cultural/ethnic effects. We additionally considered negative EFE only, preventing generalizations of our conclusions to positive

EFE. It should nevertheless be noted that the decoding of positive emotions, and centrally happiness, appears preserved in SAUD. Third, while the proportion of rejected trials (i.e., trials for which more than 50% of fixations occurred outside the face ROI) was very low in both groups, it was higher in SAUD, potentially reflecting temporary attentional lapses or reduced understanding of the instructions in this group. Fourth, although gaze measurement through eye-tracking is a gold-standard measure of overt (i.e., foveal) attentional processing, it cannot track participants' covert (i.e., peripheral) attention and hence patients might have compensated their reduced overt attention through increased covert attention. This would however not diminish the relevance of our findings since the importance of overt attention per se is established for EFE processing (Spilka et al., 2019; Vaidya et al., 2014). Fifth, our cross-sectional design prevents inferences on the causal relationships between SAUD and modified attentional EFE processing. Recent studies among individuals with familial history of alcohol use disorder and subclinical populations of adolescents and young adults presenting excessive alcohol consumption suggest that EFE recognition impairments may partly precede SAUD (Lannoy et al., 2021; Le Berre, 2019). However, it is currently impossible to determine whether EFE recognition difficulties, and their attentional correlates, are a preexisting factor and/or a consequence of SAUD. Future studies should also determine whether our results, observed in recently detoxified patients with SAUD, can be generalized to other populations presenting alcohol use disorders. Finally, our sample size did not allow for meaningful investigations of gender-contingent effects, which may be important to consider in SAUD-related social cognition research (Lewis et al., 2019).

In conclusion, we provide novel direct evidence that patients with SAUD differ from HC in their overt attention allocation during EFE processing. They centrally show reduced initial attention to informative facial features and reduced later attention to the eyes specifically. Their gaze pattern is moreover less structured than HC's. These results increase our understanding of socioaffective information processing difficulties in SAUD, confirm earlier neuroimaging studies, and may have important implications for improving social cognition and interpersonal functioning in SAUD in clinical settings.

References

- Adolphs, R. (2002). Recognizing emotion from facial expressions: Psychological and neurological mechanisms. *Behavioral and Cognitive Neuroscience Reviews*, 1(1), 21–62. <https://doi.org/10.1177/1534582302001001003>
- Adolphs, R., Gosselin, F., Buchanan, T. W., Tranel, D., Schyns, P., & Damasio, A. R. (2005). A mechanism for impaired fear recognition after amygdala damage. *Nature*, 433(7021), 68–72. <https://doi.org/10.1038/nature03086>
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders DSM-5* (5th ed.). American Psychiatric Publishing.
- Babor, T. F., Higgins-Biddle, J. C., Saunders, J. B., & Monteiro, M. G. (2001). *AUDIT: The alcohol use disorders identification test. Guidelines for use in primary care*. World Health Organization.
- Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., & Pollak, S. D. (2019). Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements. *Psychological Science in the Public Interest*, 20(1), 1–68. <https://doi.org/10.1177/1529100619832930>
- Barrett, L. F., Mesquita, B., & Gendron, M. (2011). Context in emotion perception. *Current Directions in Psychological Science*, 20(5), 286–290. <https://doi.org/10.1177/0963721411422522>
- Beck, A. T., & Steer, R. A. (1987). *Manual for the beck depression inventory*. Psychological Corp.; Harcourt Brace Jovanovich.
- Bertera, J. H., & Parsons, O. A. (1978). Impaired visual search in alcoholics. *Alcoholism: Clinical and Experimental Research*, 2(1), 9–14. <https://doi.org/10.1111/j.1530-0277.1978.tb04685.x>
- Bora, E., & Zorlu, N. (2017). Social cognition in alcohol use disorder: A meta-analysis. *Addiction*, 112(1), 40–48. <https://doi.org/10.1111/add.13486>
- Calvo, M. G., Fernández-Martín, A., Gutiérrez-García, A., & Lundqvist, D. (2018). Selective eye fixations on diagnostic face regions of dynamic emotional expressions: KDEF-dyn database. *Scientific Reports*, 8(1), Article 17039. <https://doi.org/10.1038/s41598-018-35259-w>
- Cañigueral, R., & Hamilton, A. F. D. C. (2019). The role of eye gaze during natural social interactions in typical and autistic people. *Frontiers in Psychology*, 10, Article 560. <https://doi.org/10.3389/fpsyg.2019.00560>
- Carmona-Perera, M., Clark, L., Young, L., Pérez-García, M., & Verdejo-García, A. (2014). Impaired decoding of fear and disgust predicts utilitarian moral judgment in alcohol-dependent individuals. *Alcoholism: Clinical and Experimental Research*, 38(1), 179–185. <https://doi.org/10.1111/acer.12245>
- Castellano, F., Bartoli, F., Crocamo, C., Gamba, G., Tremolada, M., Santambrogio, J., Clerici, M., & Carrà, G. (2015). Facial emotion recognition in alcohol and substance use disorders: A meta-analysis. *Neuroscience and Biobehavioral Reviews*, 59, 147–154. <https://doi.org/10.1016/j.neubiorev.2015.11.001>
- Cox, S., Bertoux, M., Turner, J. J. D., Moss, A., Locker, K., & Riggs, K. (2018). Aspects of alcohol use disorder affecting social cognition as assessed using the Mini Social and Emotional Assessment (mini-SEA). *Drug and Alcohol Dependence*, 187, 165–170. <https://doi.org/10.1016/j.drugalcdep.2018.03.004>
- Creupelandt, C., D'Hondt, F., & Maurage, P. (2019). Towards a dynamic exploration of vision, cognition and emotion in alcohol-use disorders. *Current Neuropharmacology*, 17(6), 492–506. <https://doi.org/10.2174/1570159X16666180828100441>
- Creupelandt, C., Maurage, P., Bocanegra, B., Szaffarczyk, S., de Timary, P., Deleuze, J., Lambot, C., & D'Hondt, F. (2022). Spatial frequency processing and its modulation by emotional content in severe alcohol use disorder. *Psychopharmacology*, 239(8), 2647–2657. <https://doi.org/10.1007/s00213-022-06158-w>
- Crick, N. R., & Dodge, K. A. (1994). A review and reformulation of social information-processing mechanisms in children's social adjustment. *Psychological Bulletin*, 115(1), 74–101. <https://doi.org/10.1037/0033-2909.115.1.74>
- Cuve, H. C., Castiello, S., Shiferaw, B., Ichijo, E., Catmur, C., & Bird, G. (2021). Alexithymia explains atypical spatiotemporal dynamics of eye gaze in autism. *Cognition*, 212, Article 104710. <https://doi.org/10.1016/j.cognition.2021.104710>
- D'Hondt, F., de Timary, P., Bruneau, Y., & Maurage, P. (2015). Categorical perception of emotional facial expressions in alcohol-dependence. *Drug and Alcohol Dependence*, 156, 267–274. <https://doi.org/10.1016/j.drugalcdep.2015.09.017>
- Eisenbarth, H., & Alpers, G. W. (2011). Happy mouth and sad eyes: Scanning emotional facial expressions. *Emotion (Washington, D.C.)*, 11(4), 860–865. <https://doi.org/10.1037/a0022758>
- Ekman, P., Friesen, W. V., & Hager, J. C. (2002). *Facial action coding system: Investigators guide*. Research Nexus.
- Foisy, M.-L., Kornreich, C., Petiau, C., Perez, A., Hanak, C., Verbanck, P., Pelc, I., & Philippot, P. (2007). Impaired emotional facial expression recognition in alcoholics: Are these deficits specific to emotional cues? *Psychiatry Research*, 150(1), 33–41. <https://doi.org/10.1016/j.psychres.2005.12.008>
- Freeman, C. R., Wiers, C. E., Sloan, M. E., Zehra, A., Ramirez, V., Wang, G.-J., & Volkow, N. D. (2018). Emotion recognition biases in alcohol use disorder. *Alcoholism: Clinical and Experimental Research*, 42(8), 1541–1547. <https://doi.org/10.1111/acer.13802>

- Friston, K., Adams, R., Perrinet, L., & Breakspear, M. (2012). Perceptions as hypotheses: Saccades as experiments. *Frontiers in Psychology*, 3, Article 151. <https://doi.org/10.3389/fpsyg.2012.00151>
- Gamer, M., Schmitz, A. K., Tittgemeyer, M., & Schilbach, L. (2013). The human amygdala drives reflexive orienting towards facial features. *Current Biology*, 23(20), R917–R918. <https://doi.org/10.1016/j.cub.2013.09.008>
- Gehrer, N. A., Scheeff, J., Jusyte, A., & Schönenberg, M. (2019). Impaired attention toward the eyes in psychopathic offenders: Evidence from an eye tracking study. *Behaviour Research and Therapy*, 118, 121–129. <https://doi.org/10.1016/j.brat.2019.04.009>
- Gehrer, N. A., Schönenberg, M., Duchowski, A. T., & Krejtz, K. (2018). *Implementing innovative gaze analytic methods in clinical psychology: A study on eye movements in antisocial violent offenders* [Symposium]. Proceedings of the 2018 ACM Symposium on Eye Tracking Research & Applications (pp. 1–9). <https://doi.org/10.1145/3204493.3204543>
- Günther, V., Kropidlowski, A., Schmidt, F. M., Koelkebeck, K., Kersting, A., & Suslow, T. (2021). Attentional processes during emotional face perception in social anxiety disorder: A systematic review and meta-analysis of eye-tracking findings. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 111, Article 110353. <https://doi.org/10.1016/j.pnpbp.2021.110353>
- Guo, K. (2012). Holistic gaze strategy to categorize facial expression of varying intensities. *PLoS ONE*, 7(8), Article e42585. <https://doi.org/10.1371/journal.pone.0042585>
- Hall, J. K., Hutton, S. B., & Morgan, M. J. (2010). Sex differences in scanning faces: Does attention to the eyes explain female superiority in facial expression recognition? *Cognition and Emotion*, 24(4), 629–637. <https://doi.org/10.1080/02699930902906882>
- Hoffman, L. A., Lewis, B., & Nixon, S. J. (2019). Neurophysiological and interpersonal correlates of emotional face processing in alcohol use disorder. *Alcoholism: Clinical and Experimental Research*, 43(9), 1928–1936. <https://doi.org/10.1111/acer.14152>
- Horley, K., Williams, L. M., Gonsalvez, C., & Gordon, E. (2004). Face to face: Visual scanpath evidence for abnormal processing of facial expressions in social phobia. *Psychiatry Research*, 127(1–2), 43–53. <https://doi.org/10.1016/j.psychres.2004.02.016>
- Itier, R. J., & Batty, M. (2009). Neural bases of eye and gaze processing: The core of social cognition. *Neuroscience and Biobehavioral Reviews*, 33(6), 843–863. <https://doi.org/10.1016/j.neubiorev.2009.02.004>
- Kalivas, P. W., & Volkow, N. D. (2005). The neural basis of addiction: A pathology of motivation and choice. *American Journal of Psychiatry*, 162(8), 1403–1413. <https://doi.org/10.1176/appi.ajp.162.8.1403>
- Kennedy, D. P., & Adolphs, R. (2010). Impaired fixation to eyes following amygdala damage arises from abnormal bottom-up attention. *Neuropsychologia*, 48(12), 3392–3398. <https://doi.org/10.1016/j.neuropsychologia.2010.06.025>
- Koob, G. F., & Volkow, N. D. (2010). Neurocircuitry of addiction. *Neuropsychopharmacology*, 35(1), 217–238. <https://doi.org/10.1038/npp.2009.110>
- Kordsachia, C. C., Labuschagne, I., & Stout, J. C. (2018). Visual scanning of the eye region of human faces predicts emotion recognition performance in Huntington's disease. *Neuropsychology*, 32(3), 356–365. <https://doi.org/10.1037/neu0000424>
- Kornreich, C. (2017). Commentary on Bora & Zorlu (2017): Social cognition deficits in addiction—an attachment problem? *Addiction*, 112(1), 49–50. <https://doi.org/10.1111/add.13542>
- Kornreich, C., Brevers, D., Canivet, D., Ermer, E., Naranjo, C., Constant, E., Verbanck, P., Campanella, S., & Noël, X. (2013). Impaired processing of emotion in music, faces and voices supports a generalized emotional decoding deficit in alcoholism. *Addiction*, 108(1), 80–88. <https://doi.org/10.1111/j.1360-0443.2012.03995.x>
- Kornreich, C., Petit, G., Rolin, H., Ermer, E., Campanella, S., Verbanck, P., & Maurage, P. (2016). Decoding of nonverbal language in alcoholism: A perception or a labeling problem? *Psychology of Addictive Behaviors*, 30(2), 175–183. <https://doi.org/10.1037/adb0000147>
- Krejtz, K., Duchowski, A., Szmidt, T., Krejtz, I., González Perilli, F., Pires, A., Vilaro, A., & Villalobos, N. (2015). Gaze transition entropy. *ACM Transactions on Applied Perception*, 13(1), 1–20. <https://doi.org/10.1145/2834121>
- Langner, O., Dotsch, R., Bijlstra, G., Wigboldus, D. H. J., Hawk, S. T., & van Knippenberg, A. (2010). Presentation and validation of the Radboud Faces Database. *Cognition and Emotion*, 24(8), 1377–1388. <https://doi.org/10.1080/02699930903485076>
- Lannoy, S., Duka, T., Carbia, C., Billieux, J., Fontesse, S., Dormal, V., Gierski, F., López-Caneda, E., Sullivan, E. V., & Maurage, P. (2021). Emotional processes in binge drinking: A systematic review and perspective. *Clinical Psychology Review*, 84, Article 101971. <https://doi.org/10.1016/j.cpr.2021.101971>
- Le Berre, A.-P. (2019). Emotional processing and social cognition in alcohol use disorder. *Neuropsychology*, 33(6), 808–821. <https://doi.org/10.1037/neu0000572>
- Le Berre, A.-P., Rauchs, G., La Joie, R., Segobin, S., Mézenge, F., Boudehent, C., Vabret, F., Viader, F., Eustache, F., Pitel, A.-L., & Beaunieux, H. (2013). Readiness to change and brain damage in patients with chronic alcoholism. *Psychiatry Research*, 213(3), 202–209. <https://doi.org/10.1016/j.psychres.2013.03.009>
- Lenth, R. V., Buerkner, P., Herve, M., Love, J., Miguez, F., Riebl, H., & Singmann, H. (2021). *emmeans: Estimated Marginal Means, aka Least-Squares Means* (Version 1.7.1-1) [Computer software]. <https://CRAN.R-project.org/package=emmeans>
- Lewis, B., Price, J. L., Garcia, C. C., & Nixon, S. J. (2019). Emotional face processing among treatment-seeking individuals with alcohol use disorders: Investigating sex differences and relationships with interpersonal functioning. *Alcohol and Alcoholism*, 54(4), 361–369. <https://doi.org/10.1093/alcalc/agg010>
- Liebowitz, M. R. (1987). Social phobia. *Modern Problems of Pharmacopsychiatry*, 22, 141–173. <https://doi.org/10.1159/000414022>
- MacKinnon, D. P., Lockwood, C. M., Hoffman, J. M., West, S. G., & Sheets, V. (2002). A comparison of methods to test mediation and other intervening variable effects. *Psychological Methods*, 7(1), 83–104. <https://doi.org/10.1037/1082-989X.7.1.83>
- Makris, N., Oscar-Berman, M., Jaffin, S. K., Hodge, S. M., Kennedy, D. N., Caviness, V. S., Marinkovic, K., Breiter, H. C., Gasic, G. P., & Harris, G. J. (2008). Decreased volume of the brain reward system in alcoholism. *Biological Psychiatry*, 64(3), 192–202. <https://doi.org/10.1016/j.biopsych.2008.01.018>
- Manor, B. R., & Gordon, E. (2003). Defining the temporal threshold for ocular fixation in free-viewing visuocognitive tasks. *Journal of Neuroscience Methods*, 128(1–2), 85–93. [https://doi.org/10.1016/S0165-0270\(03\)00151-1](https://doi.org/10.1016/S0165-0270(03)00151-1)
- Marinkovic, K., Oscar-Berman, M., Urban, T., O'Reilly, C. E., Howard, J. A., Sawyer, K., & Harris, G. J. (2009). Alcoholism and dampened temporal limbic activation to emotional faces. *Alcoholism: Clinical and Experimental Research*, 33(11), 1880–1892. <https://doi.org/10.1111/j.1530-0277.2009.01026.x>
- Mathôt, S., Schreij, D., & Theeuwes, J. (2012). OpenSesame: An open-source, graphical experiment builder for the social sciences. *Behavior Research Methods*, 44(2), 314–324. <https://doi.org/10.3758/s13428-011-0168-7>
- Maurage, P., de Timary, P., Tecco, J. M., Lechantre, S., & Samson, D. (2015). Theory of mind difficulties in patients with alcohol dependence: Beyond the prefrontal Cortex dysfunction hypothesis. *Alcoholism: Clinical and Experimental Research*, 39(6), 980–988. <https://doi.org/10.1111/acer.12717>
- Maurage, P., Campanella, S., Philippot, P., Vermeulen, N., Constant, E., Luminet, O., & de Timary, P. (2008). Electrophysiological correlates of the disrupted processing of anger in alcoholism. *International Journal of Psychophysiology*, 70(1), 50–62. <https://doi.org/10.1016/j.ijpsycho.2008.05.572>

- Maurage, P., de Timary, P., Billieux, J., Collignon, M., & Heeren, A. (2014). Attentional alterations in alcohol dependence are underpinned by specific executive control deficits. *Alcoholism: Clinical and Experimental Research*, 38(7), 2105–2112. <https://doi.org/10.1111/acer.12444>
- Maurage, P., Joassin, F., Philippot, P., Heeren, A., Vermeulen, N., Mahau, P., Delperdange, C., Corneille, O., Luminet, O., & de Timary, P. (2012). Disrupted regulation of social exclusion in alcohol-dependence: An fMRI study. *Neuropsychopharmacology*, 37(9), 2067–2075. <https://doi.org/10.1038/npp.2012.54>
- Maurage, P., Pabst, A., Lannoy, S., D'Hondt, F., de Timary, P., Gaudelus, B., & Peyroux, E. (2021). Tackling heterogeneity: Individual variability of emotion decoding deficits in severe alcohol use disorder. *Journal of Affective Disorders*, 279, 299–307. <https://doi.org/10.1016/j.jad.2020.10.022>
- Nummenmaa, L., Hyönä, J., & Calvo, M. G. (2006). Eye movement assessment of selective attentional capture by emotional pictures. *Emotion (Washington, D.C.)*, 6(2), 257–268. <https://doi.org/10.1037/1528-3542.6.2.257>
- O'Daly, O. G., Trick, L., Scaife, J., Marshall, J., Ball, D., Phillips, M. L., Williams, S. S., Stephens, D. N., & Duka, T. (2012). Withdrawal-associated increases and decreases in functional neural connectivity associated with altered emotional regulation in alcoholism. *Neuropsychopharmacology*, 37(10), 2267–2276. <https://doi.org/10.1038/npp.2012.77>
- Pabst, A., Heeren, A., & Maurage, P. (2020). Socio-affective processing biases in severe alcohol use disorders: Experimental and therapeutic perspectives. *Addictive Behaviors*, 106, Article 106382. <https://doi.org/10.1016/j.addbeh.2020.106382>
- Pabst, A., Maurage, P., Masson, N., & Bollen, Z. (2021). Attention allocation to emotional facial expressions in severe alcohol use disorder. <https://osf.io/gsubj>
- Pabst, A., Peyroux, E., Gautier, M., de Timary, P., & Maurage, P. (2021). Social perception and knowledge impairments in severe alcohol use disorder: Group and individual-level findings. *Drug and Alcohol Dependence*, 227, Article 109009. <https://doi.org/10.1016/j.drugalcdep.2021.109009>
- Pabst, A., Peyroux, E., Rolland, B., de Timary, P., & Maurage, P. (2020). Hostile attributional bias in severe alcohol use disorder. *Journal of Psychiatric Research*, 129, 176–180. <https://doi.org/10.1016/j.jpsychires.2020.06.031>
- Park, M.-S., Kim, S.-H., Sohn, S., Kim, G.-J., Kim, Y.-K., & Sohn, J.-H. (2015). Brain activation during processing of angry facial expressions in patients with alcohol dependency. *Journal of Physiological Anthropology*, 34(1), Article 6. <https://doi.org/10.1186/s40101-015-0046-6>
- Park, M.-S., Lee, B. H., & Sohn, J.-H. (2017). Neural substrates involved in anger induced by audio-visual film clips among patients with alcohol dependency. *Journal of Physiological Anthropology*, 36(1), Article 5. <https://doi.org/10.1186/s40101-016-0102-x>
- Philippot, P., Kornreich, C., Blairy, S., Baert, I., Den Dulk, A., Le Bon, O., Streel, E., Hess, U., Pelc, I., & Verbanck, P. (1999). Alcoholics' deficits in the decoding of emotional facial expression. *Alcoholism, Clinical and Experimental Research*, 23(6), 1031–1038. <https://doi.org/10.1111/j.1530-0277.1999.tb04221.x>
- Phillips, L. H., Channon, S., Tunstall, M., Hedenstrom, A., & Lyons, K. (2008). The role of working memory in decoding emotions. *Emotion (Washington, D.C.)*, 8(2), 184–191. <https://doi.org/10.1037/1528-3542.8.2.184>
- Quaglino, V., De Wever, E., & Maurage, P. (2015). Relations between cognitive abilities, drinking characteristics, and emotional recognition in alcohol dependence: A preliminary exploration. *Alcoholism: Clinical and Experimental Research*, 39(10), 2032–2038. <https://doi.org/10.1111/acer.12841>
- R Core Team. (2019). *R: A language and environment for statistical computing* (Version 3.6.1) [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Roelofs, K., Putman, P., Schouten, S., Lange, W.-G., Volman, I., & Rinck, M. (2010). Gaze direction differentially affects avoidance tendencies to happy and angry faces in socially anxious individuals. *Behaviour Research and Therapy*, 48(4), 290–294. <https://doi.org/10.1016/j.brat.2009.11.008>
- Rupp, C. I., Demtl, B., Osthaus, F., Kemmler, G., & Fleischhacker, W. W. (2017). Impact of social cognition on alcohol dependence treatment outcome: Poorer facial emotion recognition predicts relapse/dropout. *Alcoholism: Clinical and Experimental Research*, 41(12), 2197–2206. <https://doi.org/10.1111/acer.13522>
- Rupp, C. I., Junker, D., Kemmler, G., Mangweth-Matzek, B., & Demtl, B. (2021). Do social cognition deficits recover with abstinence in alcohol-dependent patients? *Alcoholism: Clinical and Experimental Research*, 45(2), 470–479. <https://doi.org/10.1111/acer.14537>
- Salloum, J. B., Ramchandani, V. A., Bodurka, J., Rawlings, R., Momenan, R., George, D., & Hommer, D. W. (2007). Blunted rostral anterior cingulate response during a simplified decoding task of negative emotional facial expressions in alcoholic patients. *Alcoholism: Clinical and Experimental Research*, 31(9), 1490–1504. <https://doi.org/10.1111/j.1530-0277.2007.00447.x>
- Sander, D., Grandjean, D., Kaiser, S., Wehrle, T., & Scherer, K. R. (2007). Interaction effects of perceived gaze direction and dynamic facial expression: Evidence for appraisal theories of emotion. *European Journal of Cognitive Psychology*, 19(3), 470–480. <https://doi.org/10.1080/09541440600757426>
- Schurgin, M. W., Nelson, J., Iida, S., Ohira, H., Chiao, J. Y., & Franconeri, S. L. (2014). Eye movements during emotion recognition in faces. *Journal of Vision*, 14(13), Article 14. <https://doi.org/10.1167/14.13.14>
- Setien-Ramos, I., Lugo-Marín, J., Gisbert-Gustems, L., Díez-Villoria, E., Magán-Maganto, M., Canal-Bedia, R., & Ramos-Quiroga, J. A. (2023). Eye-tracking studies in adults with autism spectrum disorder: A systematic review and meta-analysis. *Journal of Autism and Developmental Disorders*, 53(6), 2430–2443. <https://doi.org/10.1007/s10803-022-05524-z>
- Shifera, B., Downey, L., & Crewther, D. (2019). A review of gaze entropy as a measure of visual scanning efficiency. *Neuroscience & Biobehavioral Reviews*, 96, 353–366. <https://doi.org/10.1016/j.neubiorev.2018.12.007>
- Singmann, H., Bolker, B., Westfall, J., Aust, F., Ben-Shachar, M. S., Højsgaard, S., Fox, J., Lawrence, M. A., Mertens, U., Love, J., Lenth, R., & Christensen, R. H. B. (2021). *afex: Analysis of factorial experiments* (Version 1.0-1) [Computer software]. <https://CRAN.R-project.org/package=afex>
- Slidrecht, W., de Waart, R., Witkiewitz, K., & Roozen, H. G. (2019). Alcohol use disorder relapse factors: A systematic review. *Psychiatry Research*, 278, 97–115. <https://doi.org/10.1016/j.psychres.2019.05.038>
- Spielberger, C. D., & Gorsuch, R. L. (1983). *State-trait anxiety inventory for adults: Sampler set: Manual, test, scoring key*. Mind Garden.
- Spilka, M. J., Pittman, D. J., Bray, S. L., & Goghari, V. M. (2019). Manipulating visual scanpaths during facial emotion perception modulates functional brain activation in schizophrenia patients and controls. *Journal of Abnormal Psychology*, 128(8), 855–866. <https://doi.org/10.1037/abn0000468>
- Stacy, A. W., & Wiers, R. W. (2010). Implicit cognition and addiction: A tool for explaining paradoxical behavior. *Annual Review of Clinical Psychology*, 6(1), 551–575. <https://doi.org/10.1146/annurev.clinpsy.121208.131444>
- Sullivan, S., Ruffman, T., & Hutton, S. B. (2007). Age differences in emotion recognition skills and the visual scanning of emotion faces. *The Journals of Gerontology Series B: Psychological Sciences and Social Sciences*, 62(1), P53–P60. <https://doi.org/10.1093/geronb/62.1.P53>
- Toh, W. L., Rossell, S. L., & Castle, D. J. (2011). Current visual scanpath research: A review of investigations into the psychotic, anxiety, and mood disorders. *Comprehensive Psychiatry*, 52(6), 567–579. <https://doi.org/10.1016/j.comppsy.2010.12.005>
- Trick, L., Kempton, M. J., Williams, S. C. R., & Duka, T. (2014). Impaired fear recognition and attentional set-shifting is associated with brain structural changes in alcoholic patients. *Addiction Biology*, 19(6), 1041–1054. <https://doi.org/10.1111/adb.12175>
- Uekermann, J., & Daum, I. (2008). Social cognition in alcoholism: A link to prefrontal cortex dysfunction? *Addiction (Abingdon, England)*, 103(5), 726–735. <https://doi.org/10.1111/j.1360-0443.2008.02157.x>
- Vaidya, A. R., & Fellows, L. K. (2019). Ventromedial frontal lobe damage affects interpretation, not exploration, of emotional facial expressions. *Cortex*, 113, 312–328. <https://doi.org/10.1016/j.cortex.2018.12.013>

- Vaidya, A. R., Jin, C., & Fellows, L. K. (2014). Eye spy: The predictive value of fixation patterns in detecting subtle and extreme emotions from faces. *Cognition*, 133(2), 443–456. <https://doi.org/10.1016/j.cognition.2014.07.004>
- Volkow, N. D., Baler, R. D., & Goldstein, R. Z. (2011). Addiction: Pulling at the neural threads of social behaviors. *Neuron*, 69(4), 599–602. <https://doi.org/10.1016/j.neuron.2011.01.027>
- Waldthaler, J., Krüger-Zechlin, C., Stock, L., Deeb, Z., & Timmermann, L. (2019). New insights into facial emotion recognition in Parkinson's disease with and without mild cognitive impairment from visual scanning patterns. *Clinical Parkinsonism & Related Disorders*, 1, 102–108. <https://doi.org/10.1016/j.prdoa.2019.11.003>
- Wolf, R. C., Philippi, C. L., Motzkin, J. C., Baskaya, M. K., & Koenigs, M. (2014). Ventromedial prefrontal cortex mediates visual attention during facial emotion recognition. *Brain*, 137(6), 1772–1780. <https://doi.org/10.1093/brain/awu063>
- Wolf, R. C., Pujara, M., Baskaya, M. K., & Koenigs, M. (2016). Emotion recognition deficits associated with ventromedial prefrontal cortex lesions are improved by gaze manipulation. *Cortex*, 82, 255–262. <https://doi.org/10.1016/j.cortex.2016.06.017>
- Zywiak, W. H., Westerberg, V. S., Connors, G. J., & Maisto, S. A. (2003). Exploratory findings from the Reasons for Drinking Questionnaire. *Journal of Substance Abuse Treatment*, 25(4), 287–292. [https://doi.org/10.1016/s0740-5472\(03\)00118-1](https://doi.org/10.1016/s0740-5472(03)00118-1)

Received August 26, 2022

Revision received May 4, 2023

Accepted July 11, 2023 ■